How to Reliably Measure Software Performance

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[1]

5 years

~ €100M 

[2, 3]

Loose fiber optic cable that caused the measurement error [4]

How to Control Your Benchmarking Environment

How to:

1. Control your benchmarking environment.
2. Design your benchmarks.
3. Interpret benchmark results.
4. Integrate benchmarks into your workflows.

How to Control Your Benchmarking Environment

Layer	Sources of Noise	Mitigations
External	Network Temperature Vibration Noisy neighbors	Use dedicated on-prem hardware Use dedicated/bare metal cloud instances
Application	Memory layout Compilation/linking	Set up fixed builds (e.g., disable ASLR)
Kernel	Scheduling Filesystem cache	Set CPU affinity Set process priority Warm up or drop caches
CPU	Simultaneous multithreading (SMT) contention Dynamic frequency scaling (DFS)	Disable SMT Disable DFS

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Bakhvalov, "Performance Analysis and Tuning on Modern CPUs", Appendix A [5]

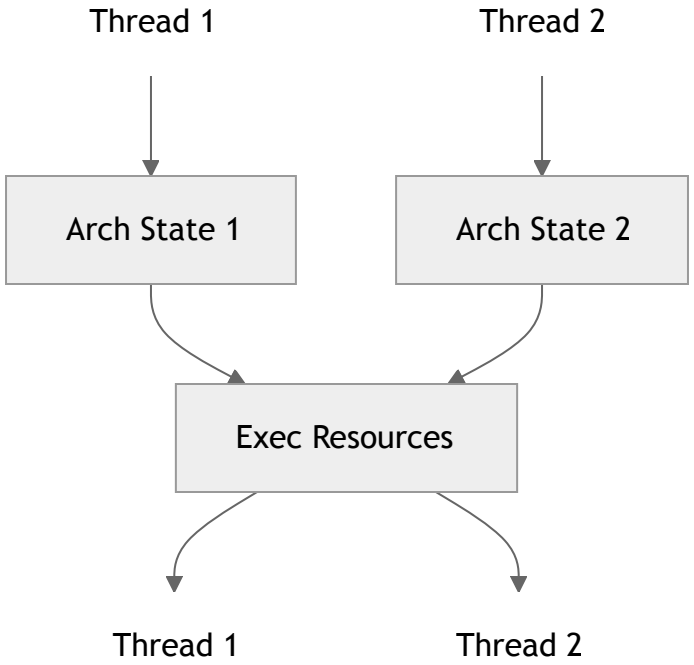
Layer	Sources of Noise	Mitigations
Kernel	Scheduling Filesystem cache	Set CPU affinity Set process priority Warm up or drop caches

```
# Set CPU affinity
taskset -c 0 ./benchmark

# Set process priority
nice -n -5 ./benchmark

# Drop filesystem cache
echo 3 > /proc/sys/vm/drop_caches && sync
```

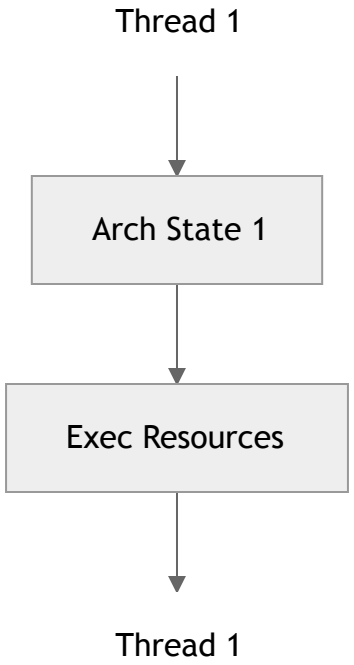
Layer	Sources of Noise	Mitigations
CPU	Simultaneous multithreading (SMT) contention	Disable SMT

```
# Disable SMT
echo off > /sys/devices/system/cpu/smt/control
```

What's SMT?



SMT enabled: hardware threads compete for resources



SMT disabled: hardware thread has exclusive access to resources

What's the impact of disabling SMT?

m5.metal, clock rate pinned, scaling governor set to "performance"

2 CPU-intensive tasks on same core (smt) vs. separate cores (no-smt)

Thread	mean ± stddev	coeff. of variation
smt-1	1537.64 ± 367.29 ms	23.887 %
smt-2	1536.88 ± 366.84 ms	23.869 %
no-smt-1	737.37 ± 0.32 ms	0.044 %
no-smt-2	737.93 ± 1.74 ms	0.235 %

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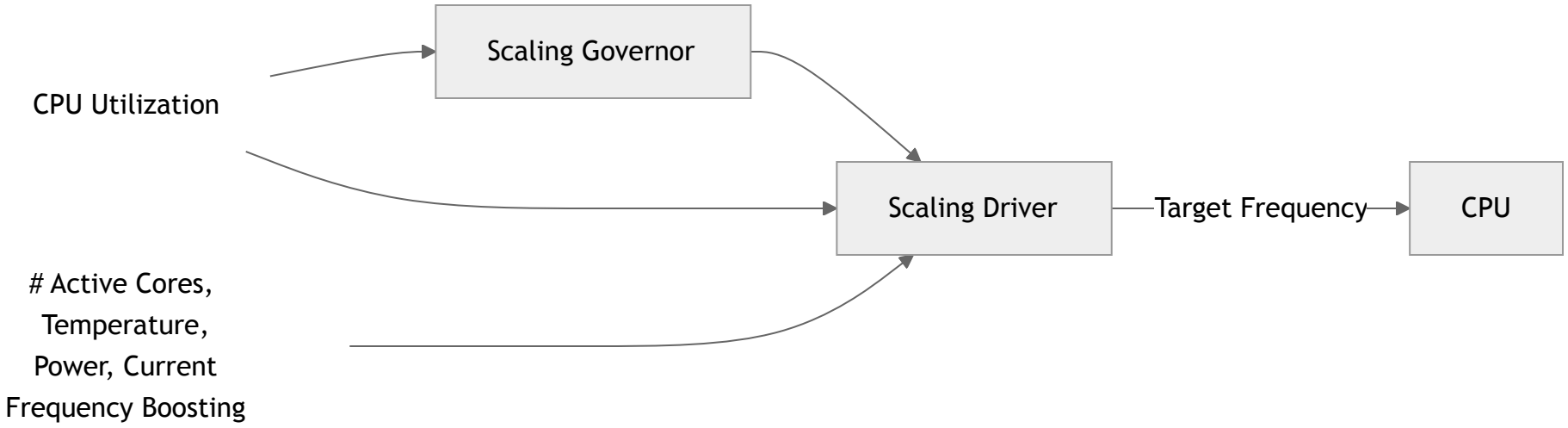
100x less variation

Layer	Sources of Noise	Mitigations
CPU	Dynamic frequency scaling (DFS)	Disable DFS

```
# Disable DFS
echo 2500000 > /sys/devices/system/cpu/cpu*/cpufreq/scaling_min_freq
echo 2500000 > /sys/devices/system/cpu/cpu*/cpufreq/scaling_max_freq
echo performance > /sys/devices/system/cpu/cpu*/cpufreq/scaling_governor

# Disable Turbo-Boost, Intel CPUs only
echo 1 > /sys/devices/system/cpu/intel_pstate/no_turbo
```

What's DFS?



DFS enabled: CPU frequency is automatically set by the Scaling Governor and the Scaling Driver

What's the impact of disabling DFS?

m5.metal, SMT disabled

Varying number of CPU-intensive tasks on the same core with DFS on vs. off

Thread	mean ± stddev	coeff. of variation
dfs-1	533.97 ± 2.046 ms	0.383 %
dfs-8	578.67 ± 0.287 ms	0.050 %
no-dfs-1	738.18 ± 0.306 ms	0.041 %
no-dfs-8	739.18 ± 0.351 ms	0.047 %

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Removes dynamic frequency scaling as a source of noise

Layer	Sources of Noise	Mitigations
External	Vibration	Don't shout in the datacenter

Shouting in the Datacenter

How to Design Benchmarks

"All happy families are alike; each unhappy family is unhappy in its own way."

— Leo Tolstoy, *Anna Karenina*

*"All happy ~~families~~ **benchmarks** are alike; each unhappy ~~family~~ **benchmark** is unhappy in its own way."*

representative and **repeatable**

An unhappy benchmark

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- System under test: Spring app instrumented (or not) with dd-trace-java.
- Workload: As many requests as possible by 5 concurrent users.
- **20 second warmup, 15 seconds of actual measurements.**

Many **false positives** and **high coeff. of variation** (= standard deviation / mean) of 11.80%.

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Are we running the benchmark long enough?

Tip #1: Run benchmarks for longer to uncover perturbations (e.g., warmup effects).

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For how long should we run the benchmark?

# measurements	coeff. of variation
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But what about inter-run variation?

Impact of initial state on FFT benchmark results [6]

Run #	mean ± stddev	coeff. of variation
1	20.08 ± 0.63 ms	3.16%
2	20.63 ± 0.56 ms	2.72%
3	20.31 ± 0.45 ms	2.23%
4	20.19 ± 0.54 ms	2.66%
5	20.26 ± 0.63 ms	3.11%
all	20.29 ± 0.60 ms	2.94%

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But what about inter-experiment variation?

Tip #0: Control your benchmarking environment.

Interpreting Benchmark Results

How can we tell if the difference is big enough?

$$\frac{\text{how big the difference is}}{\text{how big the noise is}}$$

$$t = \frac{\text{how big the difference is}}{\text{how big the noise is}}$$

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$$t > \text{critical value}$$

$$t = \frac{\text{how big the difference is}}{\text{how big the noise is}} = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

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Critical value: threshold based on your tolerance for false positives.

In practice, we use the p-value: p < α

p-value: probability of seeing this result if there's no real difference.

α: false positive rate you're willing to tolerate.

```
from scipy import stats

alpha = 0.05
t_stat, p_value = stats.ttest_ind(before, after)

if p_value < alpha:
    print("Statistically significant difference")
```

Choosing α

α = 0.05 (5%) is a common threshold for false positives.

Confidence level = 1 - α = 95%

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Trade-off

- **Lower α (1%):** Fewer false positives, fewer detections.
- **Higher α (10%):** More false positives, more detections.

Another approach: changepoint detection

Integrating Benchmarks Into Your Workflows

A series of screenshots showing the different ways in which we integrate benchmarks into our workflows at Datadog, including: a basic architecture slide, reporting capabilities, PR comments, performance quality gates, operational excellence reviews, etc.

Concluding slides

Summarize the takeaways.

References

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